

POLYTECHNIQUE MONTRÉAL

LOG6953FE - DIGITAL TWIN ENGINEERING

The brewers - DT Brewing Station

by:
Pierre-Emmanuel Goffi
&
Raphaël Tremblay

due:
16 avril 2025

Contents

1. Executive Summary	3
2. Introduction and Project Context	3
2.1. Digital Twin Concept and Application to Brewing	3
2.2. The Evolution of Brewing Technology	4
2.3. The Science of Fermentation and Monitoring Requirements	4
2.4. Challenges in Fermentation Control	5
2.5. Project Objectives and Digital Twin Classification	5
3. System Architecture and Implementation	6
3.1. Overview and Journey from Concept to Implementation	6
3.2. Physical Twin Components	7
3.2.1 Fermentation Tank Setup	7
3.2.2 Sampling Chamber	7
3.2.3 Sensor Array	7
3.2.4 Electronic Hydrometer	8
3.2.5 Arduino Control System	8
3.3. Digital Twin Components	8
3.3.1 Application Server Architecture	8
3.3.2 Database Schema	9
3.3.3 Unity Visualization System	9
3.4. Data Flow and Communication Protocols	10
3.4.1 Sensor Data Collection	10
3.4.2 Control Signal Transmission	10
3.4.3 Communication Protocols	10
3.5. DT Constellation	11
4. Process Engineering and Methodology	12
4.1. State Machine Design and Implementation	12
4.2. Safety-Critical Systems Engineering	14
4.3. Calibration and Measurement Engineering	14
5. Services and Value Delivery	14
5.1. Digital Twin Services Architecture	14
5.1.1 Visualization Service	14
5.1.2 Analysis Service	15
5.1.3 Safety Monitoring Service	15
5.1.4 Control Service	15
5.1.5 Database Service	15
5.2. Value Delivery to Stakeholders	16
6. Implementation Challenges and Solutions	16
6.1. Physical System Challenges	16
6.2. Software and Communication Challenges	17
6.3. Integration and Interdisciplinary Challenges	18
7. Status and Future Work	18
7.1. Current Implementation Status	18
7.2. Future Development Roadmap	19

7.3. Lessons Learned and Best Practices	19
8. Conclusion	20

1. Executive Summary

This report presents the design, implementation, and evaluation of our Fermentation Monitoring System, a comprehensive digital twin solution for the study of beer fermentation processes at Polytechnique Montréal (PolyBroue). The system integrates advanced sensor technology, embedded systems, and cloud architecture to create a bidirectional relationship between the physical brewing equipment and its digital representation. A demonstration video of the system can be viewed at <https://youtu.be/-EGl8y5Qlus>.

Our solution consists of two primary components:

1. A physical twin utilizing an Arduino-based control system that manages a specialized measurement chamber
2. A digital twin hosted on a Linux server that provides real-time visualization, data storage, and limited control capabilities

The physical system circulates fermentation contents through a sampling chamber, pressurizes it to 7 bar using nitrogen, and measures critical parameters including dissolved O₂, pH, conductivity, and temperature. In parallel, an electronic hydrometer continuously monitors density data directly from the fermentation tank.

Key achievements of this project include:

- The successful transformation from a digital shadow to a digital twin through the implementation of bidirectional communication protocols between the physical and digital components. This enables the digital twin to adjust non-critical parameters such as data collection frequency while maintaining robust safety constraints.
- The development of control mechanisms for the critical decompression phase, incorporating safety protocols that prevent equipment damage and operational hazards.
- The creation of a comprehensive visualization system using Unity that provides brewers with an intuitive 3D representation of the fermentation process and equipment status, complemented by a database solution that tracks parameter evolution over time.

2. Introduction and Project Context

2.1. Digital Twin Concept and Application to Brewing

A digital twin is a virtual representation of a physical object or system that spans its lifecycle, is updated from real-time data, and uses simulation, machine learning and reasoning to help decision-making [1]. In the context of brewing, this means creating a dynamic digital model that not only mirrors the physical fermentation process but also allows for bidirectional data exchange and limited control functions.

Our project represents the evolution of brewing instrumentation from:

- Traditional manual sampling and testing (providing only periodic snapshots)
- To automated data collection systems (offering continuous monitoring but limited interaction)
- To a digital twin (enabling both monitoring and controlled interaction)

This evolution addresses fundamental challenges in brewing operations: inconsistent testing due to variable staff availability and measurement techniques, limited visibility into process parameters that creates blind spots during critical fermentation phases, and the difficulty of maintaining optimal fermentation conditions amid environmental fluctuations. Traditional manual methods provide only periodic snapshots

of fermentation progress, making it challenging to identify subtle trends or rapid changes that can significantly impact beer quality. Furthermore, the complex interdependence between fermentation variables (temperature, pressure, dissolved oxygen, pH) necessitates simultaneous real-time monitoring that conventional approaches cannot provide. By implementing a digital twin with bidirectional communication capabilities, brewers gain continuous insights and limited control functions that enable more consistent product quality, improved resource efficiency, and comprehensive process documentation.

2.2. The Evolution of Brewing Technology

Beer brewing stands as one of humanity’s oldest biotechnological processes, dating back thousands of years [2]. Despite its ancient origins, modern brewing has evolved into a sophisticated operation that balances traditional craftsmanship with cutting-edge technology. The fermentation phase represents the most critical and complex stage in beer production, where yeast transforms sugars into alcohol, carbon dioxide, and various flavor compounds. The precise management of this process directly impacts product quality, consistency, and production efficiency [3].

At Polytechnique Montréal, the technical society Polybroue has developed a sophisticated fermentation monitoring prototype capable of conducting mechanical tests. However, this system previously lacked the software automation necessary to realize its full potential as a research and process control tool. The implementation of digital twin technology presented a compelling opportunity to advance both brewing science and process engineering by creating a virtual representation that not only mirrors the physical system but also enhances its capabilities through advanced analytics and limited control functions.

2.3. The Science of Fermentation and Monitoring Requirements

Beer fermentation is a dynamic biochemical process influenced by multiple variables that must be precisely monitored and controlled. During fermentation, yeast cells consume fermentable sugars and produce ethanol, carbon dioxide, and a complex array of flavor compounds. This process typically occurs over several days to weeks, depending on the beer style, with each phase requiring specific environmental conditions for optimal outcomes.

Critical parameters that must be monitored include:

- **Dissolved oxygen (O_2):** Essential for yeast health during the initial phase but detrimental to flavor stability once fermentation begins in earnest.
- **pH levels:** Indicates fermentation progress and impacts enzyme activity, yeast health, and flavor development. The typical pH decline during fermentation serves as a key indicator of process normality. If the pH drops rapidly, it can indicate the sign of bacterial contamination such as lactic acid or *Lactobacillus* bacteria. Some beers such as sours and gose also induce bacterial fermentation in their making to reduce pH, in these conditions, the readings serve as an indicator of the fermentation process.
- **Conductivity:** Provides insight into the concentration of ions and dissolved solids, which affects yeast performance and flavor compounds.
- **Temperature:** Directly influences fermentation kinetics, yeast activity, and the production of flavor compounds. Even small temperature variations can significantly alter the final product profile.
- **Density:** Measured by the electronic hydrometer, this parameter tracks the conversion of sugars to alcohol and serves as the primary indicator of fermentation completion.

Traditional monitoring methods often involve manual sampling and laboratory analysis, which introduces delays between observation and action. Furthermore, these approaches typically include only density and

temperature readings, which fail to capture the complex interactions between parameters or provide real-time insights into process dynamics. The inadequacy of conventional monitoring becomes particularly apparent when attempting to standardize production across multiple batches or when developing new beer formulations.

2.4. Challenges in Fermentation Control

Our project addresses several fundamental challenges in fermentation monitoring and control that have historically limited brewing process optimization:

Safety and Equipment Integrity: The most critical challenge involves maintaining system integrity during pressurization and decompression phases. Our monitoring device operates at pressures up to 7 bar, presenting potential hazards if not properly controlled. The digital twin must incorporate sophisticated safety protocols that prevent dangerous operating conditions while still enabling effective monitoring.

Contamination detection: In brewing, early detection of bacterial contamination or any anomalies is crucial, as delays can lead to spoiled batches and significant waste. A digital twin must monitor fermentation trends in real time, alerting brewers to potential issues so they can take timely action before it's too late.

Complex Sampling Requirements: Obtaining representative samples without disrupting the fermentation process requires precise timing and control of multiple valves, pumps, and sensors. Coordination between these components demands automated orchestration to ensure measurement accuracy and system reliability.

Data Integration and Interpretation: Correlating measurements from diverse sensors operating under different conditions (pressurized chamber vs. fermentation tank) presents significant data integration challenges. The system must synchronize and normalize these inputs to provide coherent insights into fermentation status.

Process Variability Management: Biological processes inherently exhibit variability that must be distinguished from equipment anomalies or operational issues. The digital twin must incorporate sufficient intelligence to differentiate between normal process fluctuations and actionable deviations.

Remote Monitoring with Safety Constraints: Enabling observation and limited control without compromising safety protocols requires a sophisticated bidirectional communication architecture with robust failsafe mechanisms.

2.5. Project Objectives and Digital Twin Classification

Our project aimed to transform the existing fermentation monitoring prototype into a comprehensive digital twin system that enhances brewing research capabilities while maintaining operational safety. Specifically, we intended to:

1. Develop a fully functional physical twin with automated sampling, pressurization, measurement, and decompression capabilities managed by an Arduino-based control system.
2. Create a responsive digital twin hosted on a Linux server that provides real-time visualization of system status, parameter values, and operational modes through a Unity-based interface.
3. Implement a bidirectional communication protocol that enables the digital twin to adjust non-critical parameters (such as data collection frequency) while maintaining strict safety boundaries.
4. Establish a robust database system for tracking parameter evolution throughout the fermentation process, enabling comparative analysis across multiple batches and potential integration with predictive models.

5. Incorporate sophisticated safety protocols that prevent unsafe operations, particularly during critical decompression phases, through predictive monitoring and automatic intervention.
6. Design the system architecture to support future extensions, including more advanced simulation capabilities and potential machine learning integration for predictive quality control.

In terms of digital twin classification, our system qualifies as a **Type 2 Interactive Digital Twin** according to Chaplin et al. [4], where bidirectional data flow exists and the digital representation can influence the physical system within defined safety parameters. This corresponds to a full digital twin in Kritzinger’s classification system, featuring automated data flow in both directions between the physical object and its digital counterpart.

3. System Architecture and Implementation

3.1. Overview and Journey from Concept to Implementation

The development of our digital twin system followed an evolutionary process from concept to implementation:

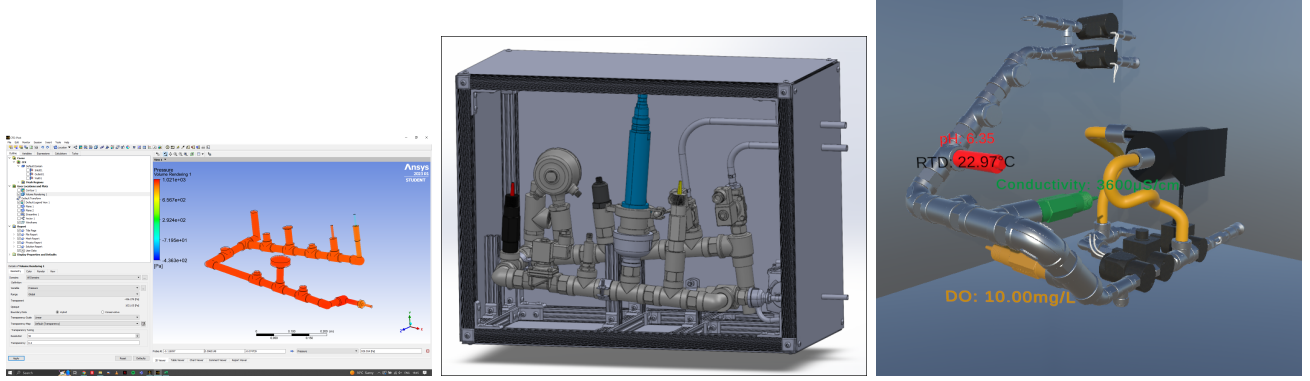


Figure 1: Evolution of the digital twin system: (a) Initial prototype sketch (Fall 2023), (b) CAD implementation (Fall 2024), and (c) Digital representation in Unity environment (Winter 2025).

Initial Prototype (Fall 2023): Design and initial mechanical implementation of the fermentation monitoring system. The conceptual foundations were established during this phase, focusing on non-invasive monitoring capabilities and safety considerations (Fig. 1a).

Mechanical Implementation (Fall 2024): Construction of the physical prototype, including fabrication of the sampling chamber and integration of sensors, valves, and the control system. This phase transformed theoretical concepts into practical engineering solutions, addressing challenges related to pressure management and sample integrity (Fig. 1b).

Digital Twin Development (Winter 2025): Implementation of the software architecture, database, visualization systems, and the bidirectional communication protocols that transform the system from a monitoring device into a digital twin. This phase included the critical translation of physical prototype CAD models into the Unity environment, requiring precise geometry conversion, texture mapping, and component relationship preservation to ensure the digital representation accurately mirrored the physical system (Fig. 1c).

Our architecture comprises two primary components: a physical twin that interfaces directly with the fermentation process and a digital twin that provides visualization, analysis, and limited control capabilities. This architecture enables real-time monitoring of critical fermentation parameters while maintaining

strict safety protocols, particularly during the pressurization and decompression phases.

3.2. Physical Twin Components

The physical twin comprises several key hardware components that interact directly with the fermentation process to collect data and control operational parameters:

3.2.1 Fermentation Tank Setup

The fermentation tank houses the primary beer fermentation process and serves as the connection point for our monitoring system. The tank is modified to accommodate:

- An outlet port for sample circulation through the monitoring system
- A return port for processed samples
- A sealed mounting point for the electronic hydrometer

The tank maintains the proper environmental conditions for fermentation while allowing our system to extract samples without compromising the sterile brewing environment.

3.2.2 Sampling Chamber

The sampling chamber represents the core of the physical twin's measurement capability and includes:

- A precision-machined stainless steel chamber rated for pressures up to 15 bar (with 10 bar operational limit)
- Input and output valves controlled by electronic relays
- A nitrogen injection system for pressure control
- A pressure release valve with precision flow control
- Sensor ports with sanitary fittings for each measurement probe

The chamber is designed to maintain sample integrity while enabling accurate measurements under controlled pressure conditions, critical for dissolved gas analysis.

3.2.3 Sensor Array

Our system incorporates multiple high-precision sensors:

- **Dissolved Oxygen (DO) Sensor:** Optical-based oxygen sensor with a detection range of 0-100 mg/L and accuracy of ± 0.05 mg/L, capable of operating under pressure
- **pH Sensor:** Industrial-grade sensor with a measurement range of 0-14 pH and accuracy of ± 0.002 pH units, featuring temperature compensation
- **Conductivity Sensor:** Four-electrode sensor with a range of 1-100,000 $\mu\text{S}/\text{cm}$ and temperature compensation
- **Temperature Sensor:** Platinum RTD (PT-1000) with accuracy of $\pm 0.01^\circ\text{C}$ across the fermentation temperature range (0-30 $^\circ\text{C}$)
- **Pressure Transducer:** Calibrated to measure 0-7 bar with ± 0.006 bar accuracy

All sensors feature digital outputs for direct communication with the Arduino controller.

3.2.4 Electronic Hydrometer

The iSpindle electronic hydrometer operates as a semi-autonomous component:

- Wireless floating hydrometer with integrated microcontroller
- Measures specific gravity (density) with ± 0.001 accuracy
- Contains internal temperature sensor for compensation
- Battery-powered with 2-week operational life
- Wi-Fi connectivity for direct data transmission to the application server

This device floats freely within the fermentation tank, providing continuous density measurements without requiring sample extraction.

3.2.5 Arduino Control System

An Arduino serves as the central controller for the physical twin:

- Controls seven relay modules that operate valves, pump, and pressure systems
- Interfaces with all sensor systems via I2C and digital I/O
- Executes the state machine that governs the sampling, pressurization, measurement, and decompression sequence
- Provides Wi-Fi connectivity through an ESP module
- Includes hardware watchdog timer for system safety
- Features custom PCB shield for reliable connections in the brewery environment

The Arduino implementation includes fail-safe programming that defaults to a safe state during any communication or power failure.

3.3. Digital Twin Components

The digital twin comprises several software components that collectively process, store, analyze, and visualize data from the physical system:

3.3.1 Application Server Architecture

The application server represents the central processing unit of the digital twin:

- Developed using Python with Flask for RESTful API endpoints
- Implements a publish-subscribe (PubSub) pattern for real-time data distribution
- Features a dedicated safety monitor module that continuously evaluates system state
- Implements authentication and authorization for secure access control
- Hosted on a Linux VPS with Docker containerization for deployment consistency

The server processes incoming sensor data, evaluates thresholds, manages operational parameters, and coordinates bidirectional communication between all system components.

3.3.2 Database Schema

A relational database (PostgreSQL) stores all system data with the following schema design:

- **fermentation_batches**: Stores metadata about each brewing session
- **measurements**: Time-series data from all sensors with timestamps and batch references
- **system_states**: Records the operational state of both physical and digital twins
- **control_commands**: Logs all control signals sent to the physical twin
- **alerts**: Documents threshold violations and system warnings
- **user_interactions**: Tracks user engagement with the digital twin

The database implements time-series optimization for efficient storage and retrieval of measurement data, with automatic aggregation for historical trend analysis.

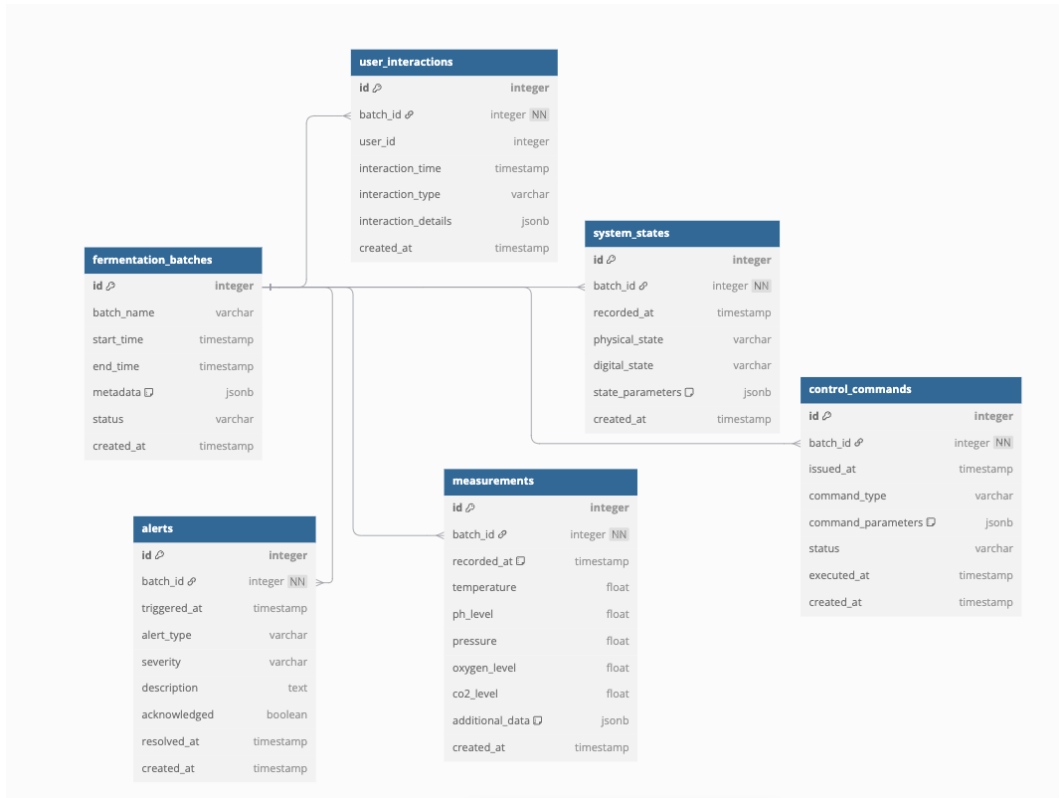


Figure 2: Database Schema with Time-Series Optimization for Digital Twin Operations

3.3.3 Unity Visualization System

Unity 3D provides an interactive visualization environment:

- Features a detailed 3D model of the physical system with real-time animation
- Displays current sensor readings with color-coded status indicators
- Includes real-time and historical graphs for trend analysis
- Provides a control interface for adjusting non-critical parameters

The visualization system serves as the primary user interface for monitoring fermentation progress and interacting with the digital twin.

3.4. Data Flow and Communication Protocols

The fermentation monitoring system employs a data flow architecture with multiple communication protocols:

3.4.1 Sensor Data Collection

Sensor data collection follows a multi-stage process designed to ensure both data integrity and system responsiveness. The Arduino controller serves as the primary data acquisition point, polling connected sensors at configurable intervals, typically every 5 seconds, with initial validation and processing occurring at the microcontroller level. In parallel, the iSpindle hydrometer independently transmits density readings via HTTP requests at 15-minute intervals, providing complementary data without requiring integration into the primary sensor array. All validated measurements are packaged in JSON format and transmitted to the application server, which applies timestamps, performs secondary validation protocols, and commits the data to persistent storage in the database. Concurrently, the system publishes real-time data to the WebSocket channel, enabling immediate visualization updates while maintaining comprehensive historical records for subsequent analysis and process optimization.

3.4.2 Control Signal Transmission

The bidirectional control functionality implements a rigorous protocol hierarchy to maintain system safety while enabling responsive operation. Control commands originate from the dashboard interface based on either direct user input. Before transmission, these commands undergo validation against predefined safety thresholds by the application server, which serves as a critical safeguard against potentially hazardous operations. Once validated, approved commands are transmitted to the Arduino controller via a secured API endpoint, with the controller programmed to acknowledge command receipt before execution to prevent timing-related errors. Upon command execution, the system reports execution status back to the application server for user notification and system status updates. All control transactions are systematically logged with precise timestamps in the database, creating an auditable record of system interventions and establishing a foundation for process refinement.

3.4.3 Communication Protocols

The system employs a carefully selected suite of communication protocols, each chosen to address specific engineering requirements and operational constraints:

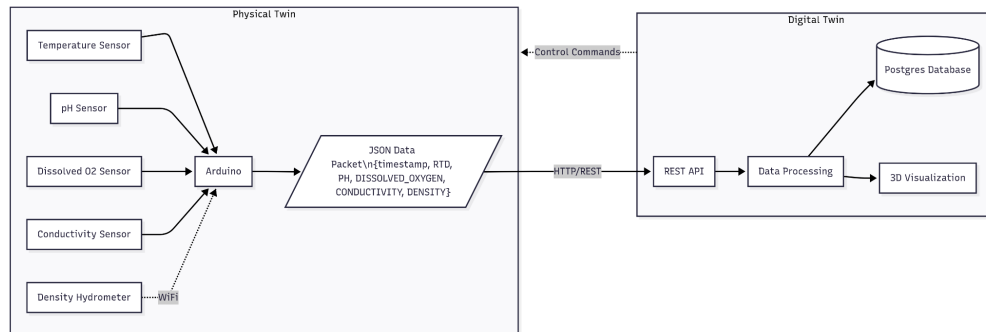


Figure 3: Data Flow Between Physical Sensors and Digital Processing Components for Fermentation Monitoring

- **HTTPS:** Implemented for all API communications to ensure data confidentiality and integrity during transmission, particularly critical given the proprietary nature of fermentation parameters and potential network vulnerabilities in academic environments.
- **WebSockets:** Selected over polling mechanisms due to its reduced latency and bandwidth efficiency, enabling near real-time data visualization required for monitoring rapid fermentation changes while minimizing server load.
- **I2C:** Adopted for microcontroller-to-sensor communications based on its simplified wiring requirements, multi-master capabilities, and reliable operation in electrically noisy brewery environments where analog signal degradation poses significant challenges.
- **Wi-Fi:** Chosen despite potential reliability concerns due to physical installation constraints that made hardwired connections impractical in the laboratory setting, with signal strength optimizations implemented to ensure consistent data transmission.
- **JSON:** Selected over binary protocols to facilitate human-readable debugging during development while maintaining compatibility with web-based visualization tools, despite the slight increase in data volume.
- **RESTful API:** Implemented to ensure scalability and facilitate future integration with additional monitoring and control systems, allowing clear separation of concerns between physical and digital components while maintaining a well-defined interface that accommodates the varying update frequencies of different sensor types.

3.5. DT Constellation

Our Digital Twin Brewing Station implements a constellation architecture that organizes the system's components into four interconnected layers: the physical brewing environment, communication infrastructure (Enablers), functional capabilities (Services), and underlying data structures (Models/Data). This layered approach clarifies the relationship between physical reality and digital representation, creating a comprehensive framework that enhances system understanding and operational visualization beyond what individual component diagrams can provide.

By visualizing our brewing Digital Twin as an integrated constellation rather than isolated components, we achieve greater coherence in system design and create a shared visual language that bridges knowledge gaps between brewing specialists, mechanical engineers, and software developers involved in the project.

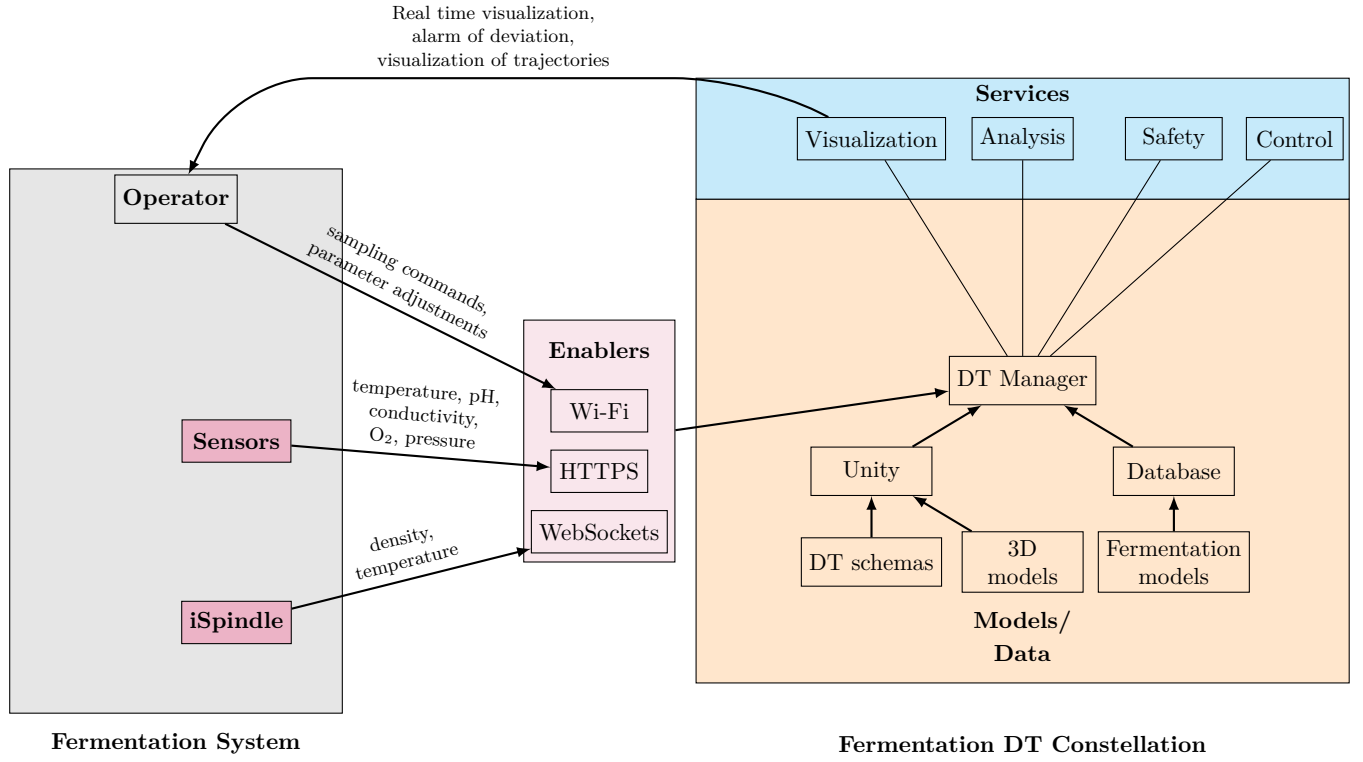


Figure 4: Constellation diagram of the brewing digital twin, showing the composition of the DT and data flow.

4. Process Engineering and Methodology

4.1. State Machine Design and Implementation

The operational logic of the physical twin is implemented as a finite state machine with the following primary states:

1. **Initial:** System at rest, waiting for trigger to begin measurement cycle
2. **FlowWay1:** First circulation path activated to draw sample from fermentation tank
3. **Pressurization:** Nitrogen injection to achieve target pressure (7 bar)
4. **Sampling:** Collection of measurement data at pressure
5. **Depressurization:** Controlled release of pressure (critical safety phase)
6. **FlowWay2:** Second circulation path activated to return sample to fermentation tank

Each state transition is governed by specific conditions (time, pressure thresholds, or external triggers) and includes appropriate safety checks. The state machine design provides a predictable operational sequence while maintaining system integrity.

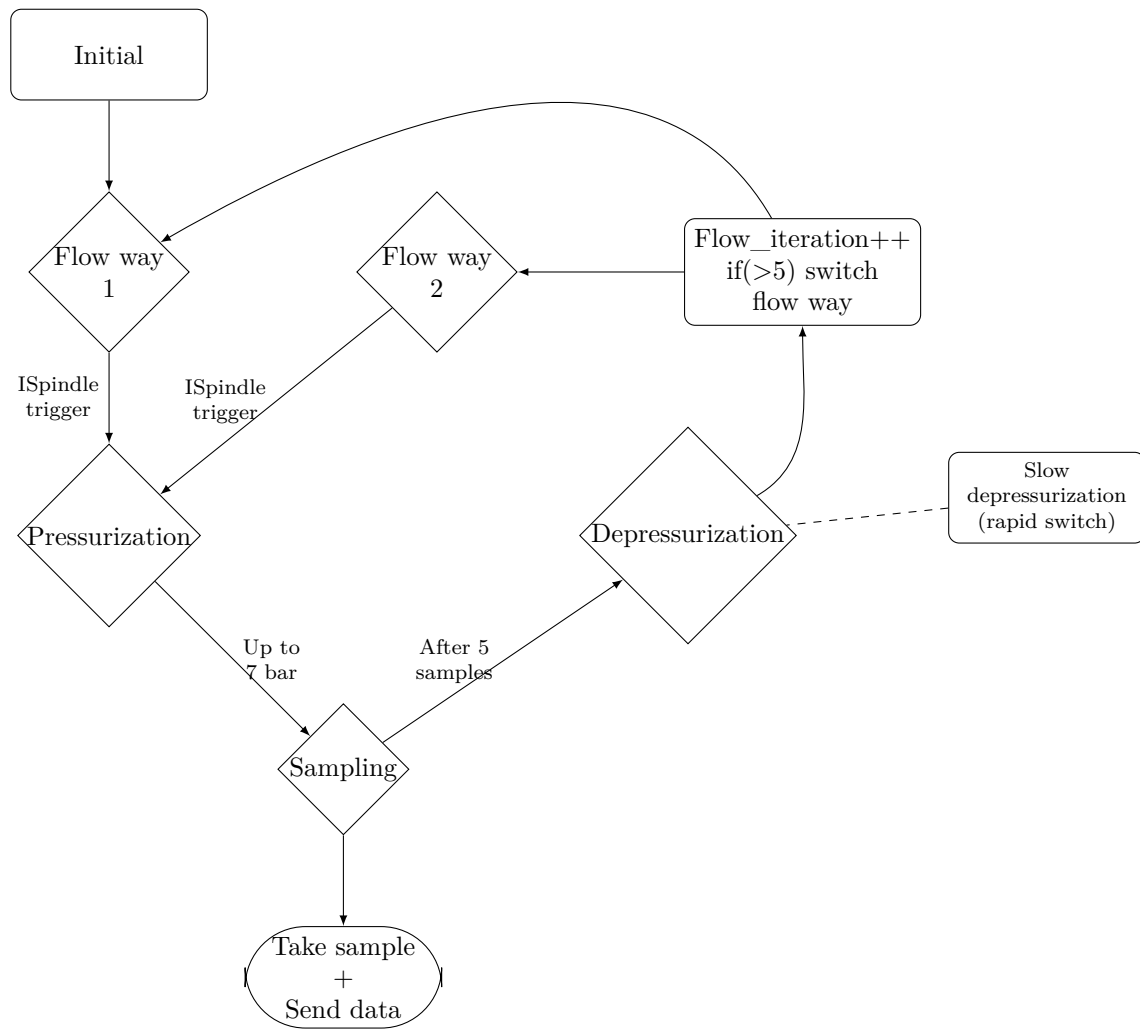


Figure 5: Flow diagram showing the sample collection and data transmission process.

4.2. Safety-Critical Systems Engineering

Given the pressurized nature of the system, safety engineering was an important consideration:

The system implements multiple layers of critical safety systems to ensure reliable operation under pressurized conditions. Software-based pressure monitoring with automatic shutdown triggers forms the first line of defense, continuously evaluating pressure conditions against safe operating parameters and autonomously initiating shutdown procedures when deviations occur. These digital protections are complemented by hardware pressure relief valves that provide mechanical failsafe functionality independent of the control system, automatically venting excess pressure regardless of software state. Watchdog timers are integrated throughout the control architecture to detect controller failures or software lockups, triggering safe state transitions when communications timeout thresholds are exceeded. Additionally, a physical emergency stop button with manual reset requirement ensures operators can immediately halt all operations in emergency situations, with the manual reset feature preventing unexpected system restart after emergency interventions.

The system’s operational framework is structured around three distinct safety levels that govern access and control permissions. Standard operations, which include non-critical parameter adjustments and monitoring functions, can be performed remotely through the digital twin interface without physical presence requirements. Limited operations introduce additional safeguards by requiring confirmation at both digital and physical interfaces before execution, creating a dual-verification protocol that prevents accidental or unauthorized activation of moderately critical functions. Critical operations, which encompass procedures with significant safety implications or equipment damage potential, are restricted to require physical presence at the brewing site, preventing any remote initiation regardless of authorization level. This hierarchical approach ensures that potentially dangerous commands cannot be executed remotely, maintaining system integrity while still allowing for remote monitoring and control of non-critical parameters that enhance operational efficiency.

4.3. Calibration and Measurement Engineering

Accurate measurement in a pressurized environment required careful calibration procedures:

Sensor Calibration Protocol: Each sensor undergoes a three-point calibration procedure under various pressure conditions to account for pressure-induced measurement deviations.

Measurement Sequencing: The sampling process follows a carefully timed sequence to ensure that:

1. Sensors are properly equilibrated before readings are recorded
2. Multiple readings are taken and averaged to reduce noise
3. Calibration corrections are applied based on current pressure and temperature

This approach ensures measurement accuracy and repeatability even under varying physical conditions.

5. Services and Value Delivery

5.1. Digital Twin Services Architecture

Our digital twin implementation follows a service-oriented architecture with five core services:

5.1.1 Visualization Service

The visualization service provides multiple interfaces for understanding the fermentation process:

- Real-time dashboard with current parameter values and trends

- Historical data visualization with configurable time ranges
- 3D Unity representation of the physical system with animated state visualization
- Alert and notification management with visual indicators

This service translates complex process data into intuitive visual representations that support decision-making by brewing technicians and researchers.

5.1.2 Analysis Service

The analysis service processes brewing data to extract actionable insights:

- Collection and storage of all sensor measurements with appropriate metadata
- Trend analysis to identify significant changes in fermentation parameters
- Comparison between actual measurements and theoretical fermentation models
- Calculation of derived metrics (e.g., fermentation rate, completion percentage)

This service transforms raw data into brewing intelligence that can inform process optimization.

5.1.3 Safety Monitoring Service

The safety service continuously evaluates system status for potential issues:

- Real-time monitoring of pressure thresholds during all operational phases
- Enforcement of physical presence requirements for critical operations
- Automatic detection of anomalous sensor readings that might indicate equipment failures
- Escalation protocols for different severity levels of safety concerns

This service ensures that the digital twin enhances rather than compromises system safety.

5.1.4 Control Service

The control service enables bidirectional interaction between digital and physical twins:

- Parameter adjustment capabilities for non-critical settings
- Data collection frequency configuration
- Safe command validation and execution
- Logging and tracking of all control operations

This service transforms the system from a passive monitoring tool to a true digital twin with bidirectional influence.

5.1.5 Database Service

The database service provides persistent storage and retrieval capabilities:

- Structured storage of all system data with appropriate relationships
- Efficient retrieval of historical data for analysis and visualization
- Time-series optimizations for performance with large datasets

- Data backup and recovery mechanisms

This service ensures that brewing knowledge is captured, preserved, and made accessible for future reference.

5.2. Value Delivery to Stakeholders

The digital twin system delivers value to multiple stakeholder groups:

Brewing Technicians:

- Reduced manual sampling requirements
- Real-time visibility into fermentation status
- Alerts for process deviations requiring attention
- Historical data for troubleshooting quality issues

Brewing Researchers:

- High-resolution data for studying fermentation dynamics
- Ability to compare actual performance with theoretical models
- Controlled environment for experimental fermentation studies
- Dataset building for future machine learning applications

Educational Value:

- Practical demonstration of digital twin concepts
- Integration of multiple engineering disciplines (mechanical, electrical, chemical, software)
- Real-world application of process control principles
- Safety-critical systems design experience

The system thus serves both practical brewing applications and educational purposes within the Polytechnique Montréal environment.

6. Implementation Challenges and Solutions

6.1. Physical System Challenges

The development of the physical twin presented several significant challenges:

Pressure System Integrity: Early prototype testing revealed leakage issues at connection points, particularly during pressure cycling. This was resolved through:

- Redesign of connection fittings with improved sealing materials
- Implementation of gradual pressure ramping rather than sudden changes
- Addition of secondary containment for connections to isolate potential leaks

Sensor Reliability Under Pressure: Initial tests showed inconsistent sensor readings under pressure conditions. This was addressed by:

- Sourcing industrial-grade sensors rated for pressure environments

- Developing pressure-specific calibration protocols
- Implementing signal filtering algorithms to reduce noise under pressure

Hardware Integration Complexity: Coordinating multiple hardware components (sensors, valves, Arduino) proved challenging. Solutions included:

- Development of a custom PCB shield to centralize connections
- Implementation of diagnostic routines for component testing
- Creation of detailed wiring documentation and maintenance procedures

Noise and Signal Interference: The brewery environment introduced electrical noise that affected sensor readings. Mitigations included:

- Installation of appropriate shielding for sensitive components
- Implementation of signal filtering at both hardware and software levels
- Relocation of certain components to reduce electromagnetic interference

Water Damage Incidents: During prototype testing, the system experienced unexpected water damage that threatened both electronic components and data integrity. This was addressed through:

- Redesign of liquid pathway sections to include overflow protection chambers that contain potential leaks
- Implementation of waterproof enclosures for vulnerable electronic components

6.2. Software and Communication Challenges

The digital twin implementation also faced several challenges:

Real-time Synchronization: Maintaining accurate synchronization between physical and digital twins required addressing:

- Network latency issues between Arduino and server
- Clock drift between system components
- Data buffering during connection interruptions

Solutions included implementing timestamp reconciliation, connection health monitoring, and data buffering with catch-up mechanisms.

3D Visualization Performance: The Unity visualization system initially faced performance issues with real-time updates. This was resolved through:

- Optimization of 3D models for web performance
- Implementation of selective update strategies rather than full-scene refreshes
- Creation of level-of-detail rendering based on user interaction

Safety-Critical Software Design: Ensuring that software failures couldn't lead to unsafe physical conditions required:

- Implementation of formal verification for critical control paths
- Development of comprehensive unit and integration testing

Database Performance with Time-Series Data: Managing large volumes of sensor data created performance challenges, addressed through:

- Implementation of time-series optimizations in PostgreSQL
- Development of data retention policies with automatic aggregation
- Creation of specialized indexes for common query patterns

6.3. Integration and Interdisciplinary Challenges

The project's interdisciplinary nature created unique challenges:

Knowledge Domain Integration: The project required expertise across multiple domains (brewing science, electrical engineering, software development, mechanical engineering, and process control). This was addressed through:

- Regular cross-disciplinary knowledge-sharing sessions
- Creation of a common vocabulary for system components and processes
- Documentation of domain-specific requirements and constraints

Testing Environment Limitations: Testing a system designed for beer fermentation presented practical challenges, addressed by:

- Development of simulators for fermentation processes
- Use of water testing for physical systems before deployment with actual beer
- Creation of accelerated test scenarios to validate long-running processes

Safety Protocol Development: Balancing remote accessibility with safety requirements required:

- Careful classification of operations into safety categories
- Implementation of physical confirmation requirements for critical operations
- Development of comprehensive testing protocols for safety systems

These challenges highlighted the complexity of implementing a true digital twin in a process engineering context, where physical systems, software, and human factors must all be carefully integrated.

7. Status and Future Work

7.1. Current Implementation Status

As of April 2025, the digital twin system has reached the following implementation milestones:

Completed Components:

- Physical-to-digital twin synchronization
- Digital twin modeling and data structure
- Real-time dashboard with historical data visualization
- 3D visualization of the prototype in Unity
- Basic safety monitoring

In-Progress Components:

- Parameter adjustment capabilities
- Safety-bounded control features
- Bug fixing and system refinement

Planned Components:

- Advanced fermentation prediction models
- Integration with broader brewery management systems

7.2. Future Development Roadmap

The project roadmap includes several planned enhancements:

Advanced Analytics Integration: Implementation of machine learning models to predict fermentation outcomes based on initial conditions and real-time measurements.

Extended Control Capabilities: Development of additional control features to adjust fermentation parameters (temperature, pressure) based on digital twin recommendations.

Quality Prediction Modeling: Integration of flavor and aroma prediction models based on fermentation parameters to support recipe development.

Multi-Tank Monitoring: Expansion of the system to monitor multiple fermentation vessels simultaneously with comparative analysis.

7.3. Lessons Learned and Best Practices

The development process yielded several important insights that will inform future digital twin implementations in brewing and other industrial applications:

Safety-First Design Philosophy: Our experience demonstrated the critical importance of integrating safety considerations at the conceptual stage rather than retrofitting them later. Early identification of pressure-related risks enabled architectural decisions that prevented potential equipment damage and personal injury. We found that collaborative risk assessment sessions involving both engineering and brewing personnel identified hazard scenarios that might have been overlooked by either discipline operating independently.

Interdisciplinary Communication: The project revealed that establishing structured communication protocols between team members with different technical backgrounds was essential for success. Weekly integration meetings with standardized reporting templates significantly improved knowledge transfer between mechanical engineers, software developers, and brewing specialists.

Progressive Implementation Strategy: Our incremental development approach—evolving from passive monitoring to active control capabilities—provided substantial benefits. This methodology allowed for validation of foundational components before introducing the complexity of bidirectional controls. The staged approach also facilitated more precise resource allocation and clearer milestone achievement assessment.

Simulation-Driven Development: Creating comprehensive digital simulations before physical integration proved invaluable for rapid iteration. Our Unity-based simulation environment enabled testing of control algorithms and user interface elements under various fermentation scenarios without risking physical equipment.

Documentation Discipline: Maintaining rigorous documentation across all system components emerged as a critical success factor. The implementation of version-controlled documentation with clear ownership

assignments ensured information consistency across mechanical, electrical, and software domains. Particularly valuable was the documentation of edge cases and failure modes, which facilitated troubleshooting and informed future enhancements.

8. Conclusion

The Fermentation Monitoring System demonstrates a successful implementation of digital twin technology in the brewing process, creating a bidirectional relationship between physical brewing equipment and its digital representation. This project exemplifies the transformation of traditional brewing into a technology-enhanced process without sacrificing the craft aspects that define quality brewing.

The system qualifies as a true digital twin through its bidirectional communication capabilities, real-time synchronization, and ability to influence the physical system within safety parameters. It represents a Type 2 Interactive Digital Twin in Chaplin’s classification, offering significant advancements in brewing instrumentation technology.

The integration challenges overcome during this project—spanning electrical engineering, software development, mechanical design, and brewing science—highlight the interdisciplinary nature of digital twin implementation in real-world applications. The safety-critical aspects of the pressurized measurement system further demonstrate the importance of rigorous engineering practices when developing systems that bridge digital and physical domains.

By providing unprecedented insights into the fermentation process while establishing a foundation for future innovations in brewing science and quality control methodologies, this project contributes to both the technical understanding of digital twins and the practical advancement of brewing technology.

References

- [1] IBM inc. (2023) What is a digital twin? [En ligne]. Disponible: <https://www.ibm.com/think/topics/what-is-a-digital-twin>
- [2] One Of Everything. (2023) Beer: The origin of civilization. [En ligne]. Disponible: <https://www.oneofeverything.ca/blog/beer-the-origin-of-civilization>
- [3] M. Josey et D. L. Maskell, “Current practices and novel developments in the fermentation of beers with non-conventional yeasts,” *Molecules*, vol. 24, n^o. 8, p. 1568, 2019.
- [4] J. Chaplin, G. Martinez-Arellano et A. Mazzoleni, “Digital twins and intelligent decision making,” dans *Digital Manufacturing for SMEs: An Introduction*, J. Chaplin, C. Pagano et S. Fort, édit. Digit-T, 2020, p. 159–186.